

```
for v in [11, 12, 15]:
    y_pred = model.predict([[v]])[0]
    y_true = v*v + 2*v + 1
    print(f"{v:2d} | {y_pred:9.2f} | {y_true:4d}")
```

The output is:

```
x | predicted | true
--+-+-----+-----
Seen (training range)
1 |      -8.00 |    4
3 |     18.00 |   16
5 |     44.00 |   36
7 |     70.00 |   64
9 |     96.00 |  100
```

Unseen (outside training range)

```
11 |    122.00 |  144
12 |    135.00 |  169
15 |    174.00 |  256
```

This is what is observed:

- The model makes noticeable errors even on training data.
- Errors increase smoothly for unseen values.
- The model cannot capture curvature.
This is a classic case of underfitting:
- The model is too simple.
- It fails to learn the true data pattern.

Overfitting example: K-Nearest Neighbors (K = 1)

K-Nearest Neighbors (KNN) makes predictions based on nearby data points. With K = 1, the model simply copies the nearest training value. The code is:

```
from sklearn.neighbors import KNeighborsRegressor
```

```
model = KNeighborsRegressor(n_neighbors=1)
```

```
# Train
model.fit(X, y)
```

```
# Predict
print("Overfitting (KNN(N=1))")
print("x | predicted | true")
print("--+-+-----+-----")
```

```
print("Seen (training range)")
for v in [1, 3, 5, 7, 9]:
    y_pred = model.predict([[v]])[0]
    y_true = v*v + 2*v + 1
    print(f"{v:2d} | {y_pred:9.2f} | {y_true:4d}")
```

```
print("\nUnseen (outside training range)")
for v in [11, 12, 15]:
    y_pred = model.predict([[v]])[0]
    y_true = v*v + 2*v + 1
    print(f"{v:2d} | {y_pred:9.2f} | {y_true:4d}")
```

The output is:

```
x | predicted | true
--+-+-----+-----
Seen (training range)
1 |      4.00 |    4
3 |     16.00 |   16
5 |     36.00 |   36
7 |     64.00 |   64
9 |    100.00 |  100
```

Unseen (outside training range)

```
11 |    121.00 |  144
12 |    121.00 |  169
15 |    121.00 |  256
```

It can be observed that:

- Predictions are perfect on training data.
- For unseen values, the model repeats the nearest known output.
- It completely fails to follow the quadratic growth.

This demonstrates overfitting:

- The model memorises training data.
- It does not learn the underlying function.

The bias–variance interpretation is:

- Underfitting corresponds to high bias and low variance. The model makes strong assumptions and is consistently wrong.
- Overfitting corresponds to low bias and high variance. The model fits training data perfectly but is unstable on new data.

The goal of machine learning is to find a balance between these two extremes as the table below indicates.

Model	Training error	Test error	Behaviour
Linear regression	High	High	Underfit
KNN (K=1)	Zero	Very High	Overfit
Ideal	Balanced	Low	Good

To conclude, underfitting means not learning enough while overfitting means learning too much. Good models learn the pattern, not the data points. **END** 🐧

By: Supriyo Ganguly

The author is a technical manager at the Electronics Corporation of India Limited, Hyderabad.